MA1521 Calculus for Computing

Lecture Group 2

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Lesson 18: Friday 16 October 2026

(Week 9)

Remark. Q17.3 to Q17.4 are leftover exercises from the Tuesday class.

Remark. Q18.1 to Q18.3 are on Section 14.8 of Thomas' Calculus.

Recall: **Change of variables (substitution) in double integrals**

absolute value

$$\iint_R f(x,y)\,dx\,dy = \iint_G f(x(u,v),y(u,v))\left|\frac{\partial(x,y)}{\partial(u,v)}\right|du\,dv.$$

Or alternatively,

$$\iint_R f(x,y)\,dx\,dy = \iint_G f(x(u,v),y(u,v))\left|\frac{\partial(u,v)}{\partial(x,y)}\right|^{-1}du\,dv.$$

Here R is a region in the xy-plane, and G is the corresponding region in the uv-plane. Also,

$$\frac{\partial(x,y)}{\partial(u,v)} \overset{\substack{x\\ y}}{=} \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\cdot\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\cdot\frac{\partial y}{\partial u}.$$

Q17.3. Evaluate the integral $\iint_R (x^2 + 2xy)\, dA$ where R is the region bounded by the lines $y = 2x + 3$, $y = 2x + 1$, $y = 5 - x$ and $y = 2 - x$.

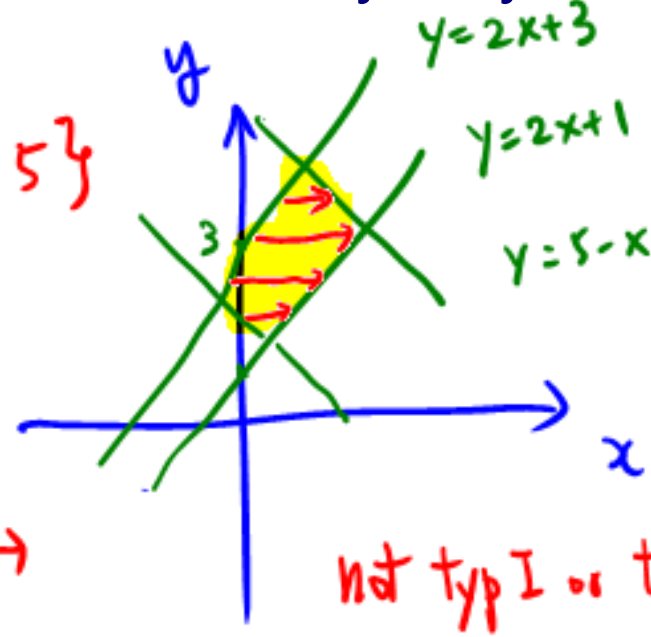
[Suggestion: From the equations of the boundary lines of R, get a transformation $u = u(x, y)$, $v = v(x, y)$, so that the region G in the uv-plane will be described by very simple inequalities involving u, v.]

$R = \{(x,y) : 1 \le y - 2x \le 3,\ 2 \le x + y \le 5\}$

Let $v = y - 2x$, $u = x + y$.

$G = \{(u,v) : 1 \le v \le 3,\ 2 \le u \le 5\}$





not typ I or type II

Solution: Note that

$$R = \{\,(x, y) \,:\, 2 \le x + y \le 5,\ \ 1 \le y - 2x \le 3\,\}$$

We let

$$\begin{cases} u = x + y \\ v = y - 2x \end{cases} \quad \text{or equivalently,} \quad \begin{cases} x = \dfrac{u - v}{3} \\ y = \dfrac{2u + v}{3} \end{cases} . \qquad (1)$$

Then from (1), we have

$$\begin{cases} 2 \le x + y \le 5 \\ 1 \le y - 2x \le 3 \end{cases} \iff \begin{cases} 2 \le u \le 5 \\ 1 \le v \le 3 \end{cases} . \qquad (2)$$

Thus the corresponding region in the uv-plane is:

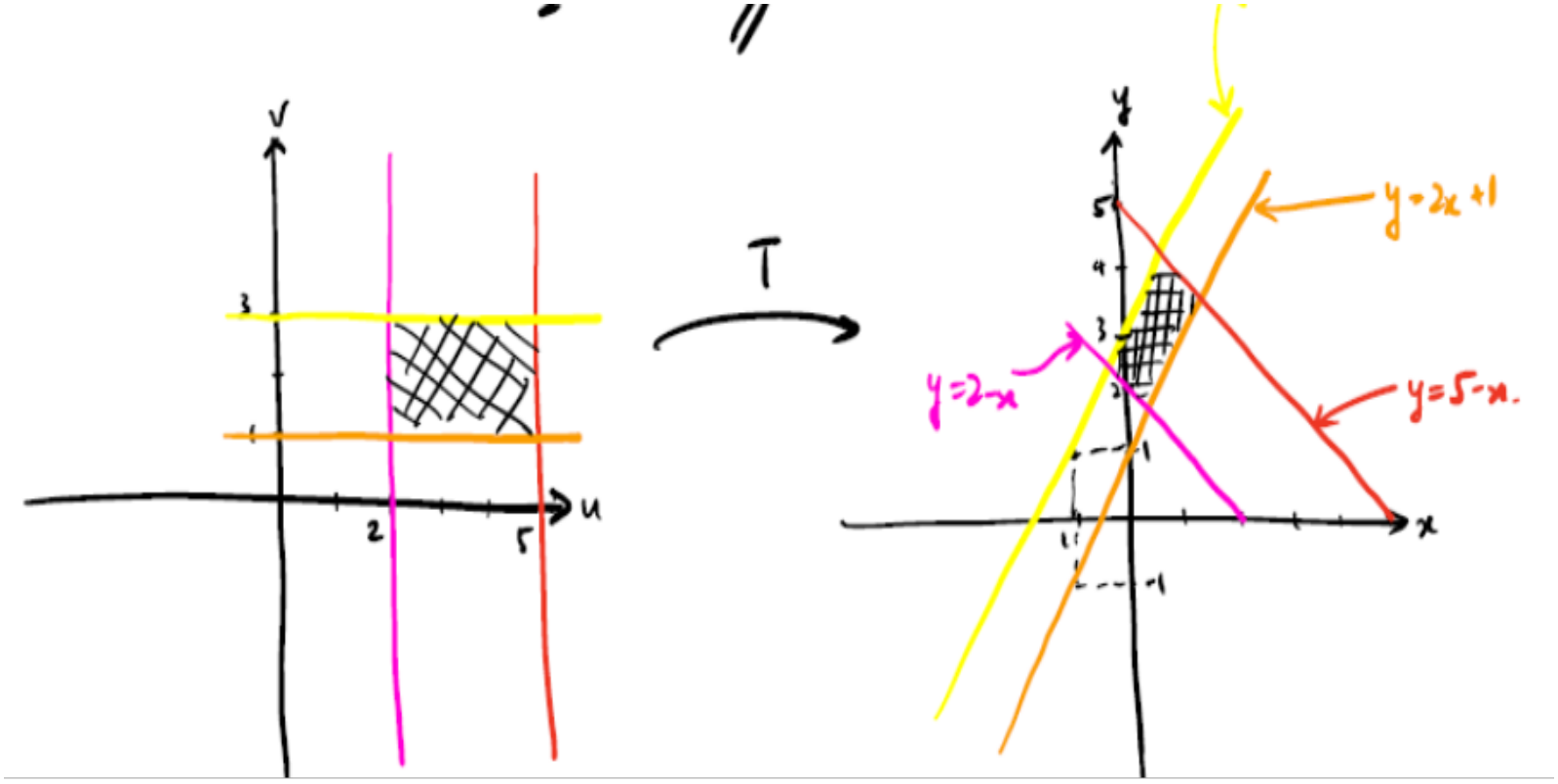
$$G = \{(u, v) \,:\, 2 \le u \le 5,\ \ 1 \le v \le 3\}.$$

$$\begin{cases} v = y - 2x & -(1) \\ u = x + y & -(2) \end{cases}$$

$(2)-(1)$: $u - v = 3x \Rightarrow x = \frac{u-v}{3}$

$(1)+(2)\times 2$: $v + 2u = 3y \Rightarrow y = \frac{v+2u}{3}$

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{matrix} x \\ y \end{matrix}\begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} \begin{matrix} \\ \\ \end{matrix} = \begin{matrix} u & v \\ \end{matrix}\begin{vmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix} = \frac{1}{9} + \frac{2}{9} = \frac{1}{3}.$$



Now

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & -\frac{1}{3} \\ \frac{2}{3} & \frac{1}{3} \end{vmatrix} = \frac{1}{9} - (-\frac{2}{9}) = \frac{1}{3}.$$

From the change of variables formula for double integrals, we have

$$x = \frac{u-v}{3}$$

$$y = \frac{2u+v}{3}$$

$$\iint_R (x^2 + 2xy)\, dA$$

$$= \iint_R (x^2 + 2xy)\, dx\, dy$$

$$= \iint_G \left(\left(\frac{u-v}{3}\right)^2 + 2\left(\frac{u-v}{3}\right)\left(\frac{2u+v}{3}\right) \right) \left| \frac{\partial(x,y)}{\partial(u,v)} \right| du\, dv$$

$$= \int_1^3 \int_2^5 \frac{(u^2 - 2uv + v^2) + (4u^2 - 2uv - 2v^2)}{9} \cdot \frac{1}{3}\, du\, dv$$

$$= \int_1^3 \left(\int_2^5 \frac{5u^2 - 4uv - v^2}{27}\, du \right) dv$$

$$= \frac{1}{27} \int_1^3 \left[\frac{5u^3}{3} - 2u^2 v - v^2 u \right]_2^5 dv$$

$$= \frac{1}{27}\int_1^3 \left(\frac{625}{3} - 50v - 5v^2 - \frac{40}{3} + 8v + 2v^2\right) dv$$
$$= \frac{1}{27}\int_1^3 \left(\frac{585}{3} - 42v - 3v^2\right) dv$$
$$= \frac{1}{27}\left[\frac{585}{3}v - 21v^2 - v^3\right]_1^3$$
$$= \frac{1}{27}\left(585 - 189 - 27 - \frac{585}{3} + 21 + 1\right) = \frac{196}{27}.$$

Q17.4. Evaluate $\iint_\Omega (x^2 + y^2)\, dA$, where Ω is the region in the xy-plane bounded by the lines

$$x + y = 0, \quad x + y = 1, \quad 2x - y = 0, \quad 2x - y = 1.$$

[Suggestion: Make a change of variables.]

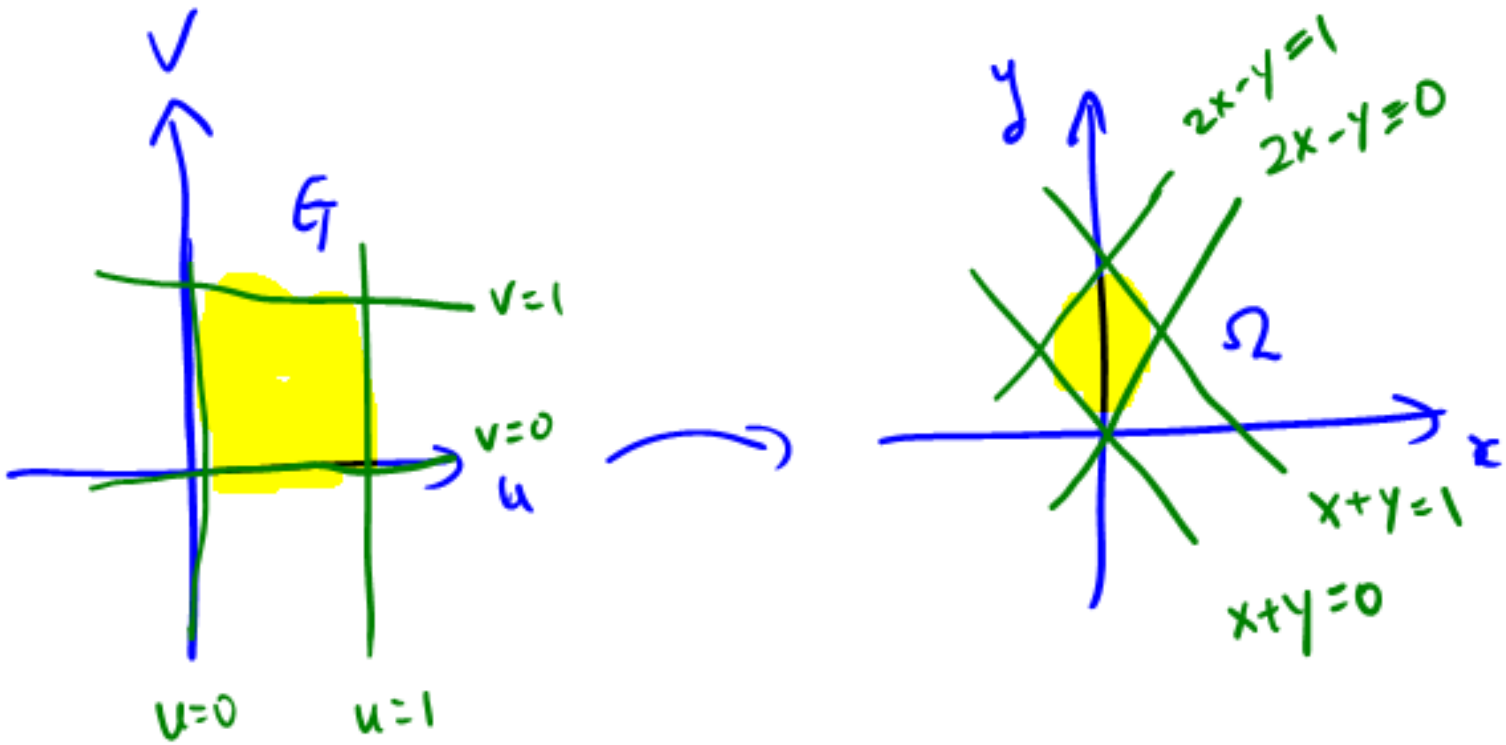
Let $u = x+y$, $v = 2x-y$. $\Rightarrow \begin{cases} x = \frac{u+v}{3}, \\ y = \frac{2u-v}{3}. \end{cases}$

$\Omega = \{(x,y) : 0 \le x+y \le 1, \ 0 \le 2x-y \le 1\}$

$G = \{(u,v) : 0 \le u \le 1, \ 0 \le v \le 1\}$

Solution: Let $u = x + y$ and $v = 2x - y$. Note that

$$\begin{cases} u = x + y \\ v = 2x - y \end{cases} \iff \begin{cases} x = \frac{1}{3}u + \frac{1}{3}v \\ y = \frac{2}{3}u - \frac{1}{3}v \end{cases} \qquad (1)$$



$$\begin{cases} u = x + y \\ v = 2x - y \end{cases} \iff \begin{cases} x = \frac{1}{3}u + \frac{1}{3}v \\ y = \frac{2}{3}u - \frac{1}{3}v \end{cases} \qquad (1)$$

By (1), we have

$$x + y = 0 \iff u = 0,$$

$$x + y = 1 \iff u = 1,$$

$$2x - y = 0 \iff v = 0,$$

$$2x - y = 1 \iff v = 1.$$

Thus the region in the uv-plane corresponding to Ω is:

$$G = \{(u, v) \ : \ 0 \leq u \leq 1, \ \ 0 \leq v \leq 1\}.$$

$$x = \frac{u+v}{3}, \quad y = \frac{2u-v}{3}$$

Now

$$\frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{2}{3} & -\frac{1}{3} \end{vmatrix} = -\frac{1}{9} - \left(\frac{2}{9}\right) = -\frac{1}{3}.$$

From the change of variables formula for double integrals, we have

$$\iint_{\Omega} (x^2 + y^2)\, dA$$

$$= \iint_{R} (x^2 + y^2)\, dx\, dy$$

$$= \iint_{G} \left((\frac{1}{3}u + \frac{1}{3}v)^2 + (\frac{2}{3}u - \frac{1}{3}v)^2 \right) \left| \frac{\partial(x, y)}{\partial(u, v)} \right| du\, dv$$

$$= \frac{1}{9} \int_0^1 \int_0^1 \left((u + v)^2 + (2u - v)^2 \right) \cdot \left| -\frac{1}{3} \right| du\, dv$$

$= \frac{1}{3}$

$$= \frac{1}{27} \int_0^1 \int_0^1 \left(u^2 + 2uv + v^2 + 4u^2 - 4uv + v^2 \right) du\, dv$$

$$= \frac{1}{27} \int_0^1 \int_0^1 \left(5u^2 - 2uv + 2v^2 \right) du\, dv$$

$$= \frac{1}{27} \int_0^1 \left[\frac{5}{3} u^3 - u^2 v + 2 v^2 u \right]_0^1 dv$$

$$= \frac{1}{27} \int_0^1 \left(\left(\frac{5}{3} - v + 2v^2 \right) - (0 - 0 + 0) \right) dv$$

$$= \frac{1}{27} \int_0^1 \left(\frac{5}{3} - v + 2v^2 \right) dv$$

$$= \frac{1}{27} \left[\frac{5}{3} v - \frac{1}{2} v^2 + \frac{2}{3} v^3 \right]_0^1$$

$$= \frac{1}{27} \left(\left(\frac{5}{3} - \frac{1}{2} + \frac{2}{3} \right) - (0 - 0 + 0) \right)$$

$$= \frac{1}{27} \cdot \frac{11}{6} = \frac{11}{162}.$$

Q18.1. Evaluate $\iint_\Omega xy\,dx\,dy$, where Ω is the region in the first quadrant ($x \geq 0,\ y \geq 0$) bounded by the curves

$$x^2+y^2=4, \quad x^2+y^2=9, \quad x^2-y^2=1, \quad x^2-y^2=4.$$

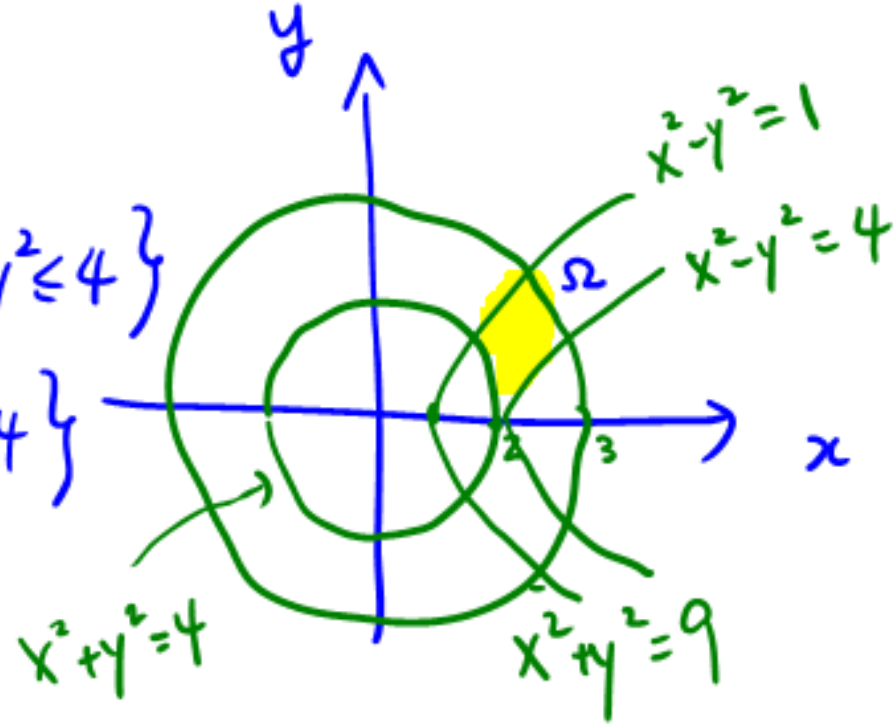
[Suggestion: Make a change of variables.]

Let $u = x^2+y^2$, $v = x^2-y^2$

$\Omega = \{(x,y) : 4 \leq x^2+y^2 \leq 9,\ 1 \leq x^2-y^2 \leq 4\}$

$G = \{(u,v) : 4 \leq u \leq 9,\ 1 \leq v \leq 4\}$





Solution: Let $u = x^2 + y^2$ and $v = x^2 - y^2$. Note that for $x \geq 0$ and $y \geq 0$,

$$\begin{cases} u = x^2 + y^2 \\ v = x^2 - y^2 \end{cases} \iff \begin{cases} x = \sqrt{\dfrac{u+v}{2}} \\ y = \sqrt{\dfrac{u-v}{2}} \end{cases} \qquad (1)$$

$u + v = 2x^2$

$\Rightarrow \quad x = \sqrt{\dfrac{u+v}{2}}$

$\dfrac{\partial(x,y)}{\partial(u,v)}$ is little complicated!

$x \geq 0$

$y \geq 0$

$$\frac{\partial(u,v)}{\partial(x,y)} \overset{\begin{matrix} x & y \\ u & \\ v & \end{matrix}}{=} \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & 2y \\ 2x & -2y \end{vmatrix} = -8xy .$$

$$\begin{cases} u = x^2 + y^2 \\ v = x^2 - y^2 \end{cases} \iff \begin{cases} x = \sqrt{\dfrac{u+v}{2}} \\ y = \sqrt{\dfrac{u-v}{2}} \end{cases} \tag{1}$$

By (1), we have

$$x^2 + y^2 = 4 \iff u = 4,$$

$$x^2 + y^2 = 9 \iff u = 9,$$

$$x^2 - y^2 = 1 \iff v = 1,$$

$$x^2 - y^2 = 4 \iff v = 4.$$

Thus the region in the uv-plane corresponding to Ω is:

$$G = \{(u, v) \ : \ 4 \leq u \leq 9, \ \ 1 \leq v \leq 4\}.$$

Now

$$\frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} \end{vmatrix} = \begin{vmatrix} 2x & 2y \\ 2x & -2y \end{vmatrix} = -4xy - 4xy = -8xy.$$

From the change of variables formula for double integrals, we have

$$\iint_\Omega xy\,dx\,dy$$

$$= \iint_G xy \left|\frac{\partial(u,v)}{\partial(x,y)}\right|^{-1} du\,dv$$

$$= \iint_G xy\,|-8xy|^{-1}\,du\,dv \qquad x, y \geq 0$$

$$= \iint_G \cancel{xy} \cdot \frac{1}{8\cancel{xy}}\,du\,dv$$

$$= \frac{1}{8}\iint_G du\,dv$$

$$= \frac{1}{8}\int_1^4\int_4^9 du\,dv = \frac{1}{8}\int_1^4 \Big[u\Big]_4^9\,dv = \frac{1}{8}\int_1^4 (9-4)\,dv$$

$$= \frac{1}{8}\int_1^4 5\,dv = \frac{1}{8}\Big[5v\Big]_1^4 = \frac{1}{8}(5\cdot 4 - 5\cdot 1) = \frac{15}{8}.$$

Q18.2. Recall that to compute double integrals using polar coordinates, we have

$$\iint_R f(x, y)\, dA = \iint_G f(r\cos\theta, r\sin\theta)\, r\, dr\, d\theta. \qquad (1)$$

Here R is a region in the xy-plane, and G is the corresponding region in the $r\theta$-plane. Use the change of variables formula for double integrals to derive (1).

$dx\,dy \rightsquigarrow r\,dr\,d\theta$

Solution:

By the change of variables formula for double integrals, we have

$$\iint_R f(x,y)\,dA = \iint_R f(x,y)\,dx\,dy$$
$$= \iint_G f(r\cos\theta, r\sin\theta)\left|\frac{\partial(x,y)}{\partial(r,\theta)}\right| dr\,d\theta.$$

Then (1) follows immediately from the following computation:

$$X = r\cos\theta, \quad Y = r\sin\theta$$

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{matrix} x \\ y \end{matrix}\begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} \quad \begin{matrix} r & \theta \end{matrix}$$

$$= \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = \cos\theta\cdot r\cos\theta - (-r\sin\theta)\cdot\sin\theta$$

$$= r(\cos^2\theta + \sin^2\theta) = r.$$

Recall that $x = r\cos\theta$, $y = r\sin\theta$. So we have

$$\frac{\partial(x,y)}{\partial(r,\theta)} = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix}$$

$$= \cos\theta \cdot (r\cos\theta) - (-r\sin\theta)\sin\theta$$

$$= r(\cos^2\theta + \sin^2\theta) = r \cdot 1 = r.$$

Q18.3. Evaluating the following iterated integral by converting to polar coordinates and using the change of variables formula for double integrals:

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} xy \, dy \, dx.$$

$x \geq 0$
$y \geq 0$

$$R = \{(x,y) : 0 \leq x \leq 2, \quad 0 \leq y \leq \sqrt{2x-x^2}\}$$

Let $x = r\cos\theta$, $y = r\sin\theta$.

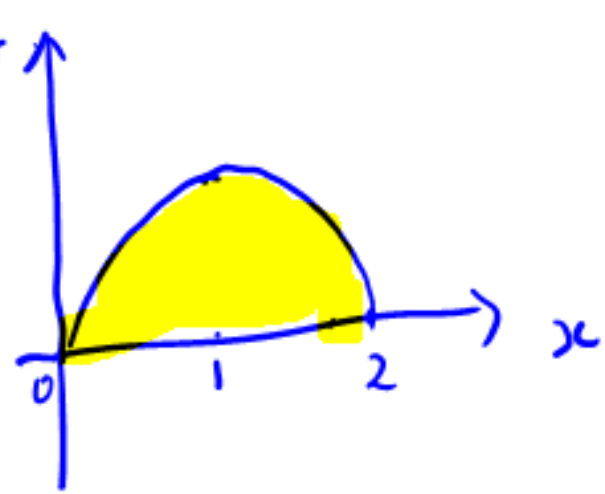
For the bounding curve, for $r > 0$,

$$y = \sqrt{2x-x^2}$$

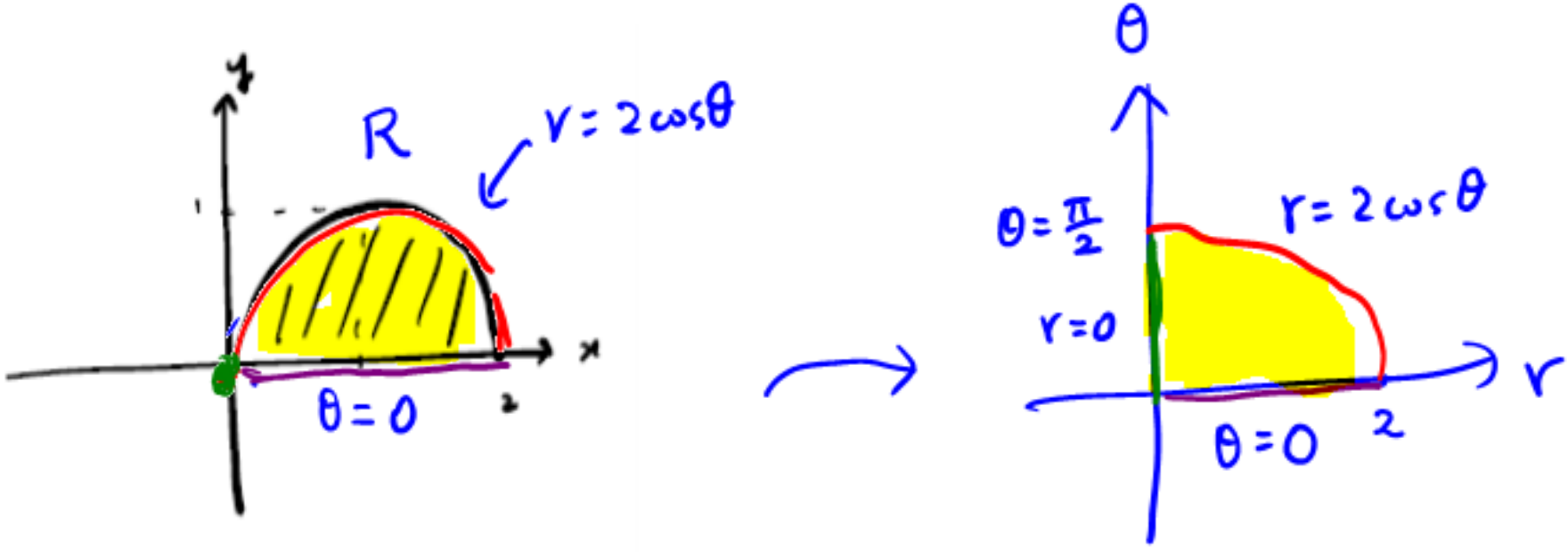
$$y^2 = 2x - x^2$$

$$\Rightarrow \quad x^2 + y^2 = 2x \Rightarrow r^2 = 2r\cos\theta \Rightarrow r = 2\cos\theta$$

$$(x-1)^2 + y^2 = 1$$



Solution:



In polar coordinates (r, θ), we have

$$\begin{aligned} y = \sqrt{2x - x^2} &\iff x^2 + y^2 - 2x = 0 \\ &\iff r^2 - 2r\cos\theta = 0 \\ &\iff r(r - 2\cos\theta) = 0 \\ &\iff r = 0 \text{ or } r = 2\cos\theta. \end{aligned}$$

Thus corresponding to the region

$$D = \left\{ (x, y) \ : \ 0 \le x \le 2, \ \ 0 \le y \le \sqrt{2x - x^2} \right\}$$

in the xy-plane is the region

$$G = \left\{ (r, \theta) \ : \ 0 \le \theta \le \frac{\pi}{2}, \ \ 0 \le r \le 2\cos\theta \right\}$$

in the $r\theta$-plane.

Tricky variable: r
Simple variable: θ

Recall that $x = r\cos\theta$, $y = r\sin\theta$. So we have

$$\frac{\partial(y, x)}{\partial(r, \theta)} = \begin{vmatrix} \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \\ \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \sin\theta & r\cos\theta \\ \cos\theta & -r\sin\theta \end{vmatrix}$$

$$= \sin\theta \cdot (-r\sin\theta) - (r\cos\theta) \cdot \cos\theta$$

$$= -r(\sin^2\theta + \cos^2\theta) = -r \cdot 1 = -r.$$

Then by the change of variables formula for double integrals, we have

$$\int_0^2 \int_0^{\sqrt{2x-x^2}} xy \, dy \, dx = \iint_D xy \, dy \, dx$$

$$= \iint_G (r\cos\theta)(r\sin\theta) \left| \frac{\partial(y,x)}{\partial(r,\theta)} \right| dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \int_0^{2\cos\theta} r^2 \cos\theta \sin\theta \cdot |-r| \, dr \, d\theta$$

$$= \int_0^{\frac{\pi}{2}} \left(\int_0^{2\cos\theta} r^3 \cos\theta \sin\theta \, dr \right) d\theta$$

$$= \int_0^{\frac{\pi}{2}} \cos\theta \sin\theta \left[\frac{r^4}{4} \right]_0^{2\cos\theta} d\theta$$

$$= 4 \int_0^{\frac{\pi}{2}} \cos^5\theta \sin\theta \, d\theta = 4 \left[-\frac{\cos^6\theta}{6} \right]_0^{\frac{\pi}{2}} = -\frac{2}{3}(0-1) = \frac{2}{3}.$$

$u = \cos\theta$

$\frac{du}{d\theta} = -\sin\theta$

$-du = \sin\theta \, d\theta$